Veer Kakar

veerkakar1@gmail.com | linkedin.com/in/VeerKakar | github.com/VeerKakar17 | PortfolioWebsite

Education

University of Pennsylvania — Philadelphia, PA
BSE in Computer Science | GPA: 3.99/4.0 — *Aug. 2024 – May 2028*

University of Pennsylvania — Philadelphia, PA
MSE in Computer Science, Systems Concentration | GPA: 4.0/4.0 — *Jan. 2026 – May 2028*
Relevant Coursework: Distributed Systems, GPU Programming, Databases, Network Security, Algorithms, OS

Experience

Software Engineering Intern — May 2026 – Present
Roblox — *San Mateo, CA*
- Developed new content review workflow, saving partners at Netflix, Kodansha, and Lionsgate 2+ hours weekly.
- Built dismissal and feedback systems to train our ML deep-scan system to identify irrelevant content before it reaches our partners' content review queues.
- Defined service boundaries, schemas, RPC contracts, consistency behavior, and asynchronous execution.
- Created durable Temporal workflows with persisted state and automatic recovery from partial failures.
- Owned two IP-licensing launches from design through production, authoring three technical specifications and coordinating implementation across platform, data science, product, and design teams.

Software Engineering Intern — May 2025 – August 2025
TransformaTech, Inc - an IBM Business Partner — *Exton, PA*
- Designed & developed a scalable Java application for automatically generating IBM App Connect Enterprise applications, significantly accelerating production and enabling faster client-specific customization.
- Integrated enterprise applications for clients (Humana, SMBC), ensuring scalable data pipelines & reliable systems.
- Improved throughput with asynchronous data exchange between heterogeneous systems using IBM MQ and Kafka.

Operating Systems and Data Structures & Algorithms Teaching Assistant — May 2025 – Present
University of Pennsylvania — *Philadelphia, PA*
- Selected to concurrently TA Operating Systems and DSA, supporting 350+ students across both courses.
- Lead weekly 20-student DSA sections and 150+ student OS recitations; hold course-wide office hours.
- Develop and grade exams, proofs, programming assignments, and systems projects; conduct oral exams and evaluate project demos through system-design questioning.

Associate App Developer Intern — May 2024 – August 2024
Aetna Inc, a CVS Health Company — *Blue Bell, PA*

Projects

Bare-Metal Raspberry Pi 5 Operating System | *C, ARMv8-A, QEMU, Unix/POSIX* — May 2026 – Present
- Built a bare-metal AArch64 Unix-style OS for RPi 5/QEMU with user/kernel isolation and tabbed GUI TTYs.
- Reverse-engineered sparse RPi 5 docs via Linux source to implement memory-mapped drivers and hardware I/O.
- Implemented per-process virtual memory with demand paging, lazy allocation, mmap, and copy-on-write fork.
- Built ext2-style inode VFS and ELF loader; added POSIX signals, pipes, shell, threads, and synchronization.

Mini Minecraft | *C++, Multithreading, OpenGL, Real-Time Rendering, Optimization* — February 2025 – May 2025
- Developed a high-performance Minecraft recreation in C++/OpenGL with a focus on systems-level optimization.
- Implemented multithreaded terrain generation and face-culling to reduce rendering overhead by over 60%.
- Designed custom redstone logic system, custom block VBOs, flowing water, and shadow mapping from scratch.

Penn Mobile Android Developer | *Kotlin, Android, CI/CD* — January 2026 – Present
- Collaborate with Design and Product to maintain Penn Mobile, a production Android app with 5,000+ downloads.
- Migrated core features from deprecated XML frameworks to Jetpack Compose, improving performance by 15%.

Technical Skills

Languages: C, C++, Rust, Python, Java, AArch64 Assembly, SQL, TypeScript, C#, Kotlin, OCaml
Systems & Tools: Linux/Unix, Virtual Memory, Processes, Threads, Filesystems, POSIX, Networking, GDB, CMake
GPU & Performance: CUDA, GPU Programming, OpenGL, Vulkan, Multithreading, Profiling, Optimization
Distributed & Cloud: gRPC, Temporal, Kafka, IBM MQ, Fault Tolerance, AWS, Docker, PostgreSQL, S3
Interests: Crocheting, Jazz Drumming, Tennis, Taekwondo (3rd degree black belt)